

Develop a data dictionary for this dataset. For each attribute:

- Determine the data type
- Determine the number of unique values
- Determine the range of numeric values
- Enumerate categorical or ordinal values
- Note any inconsistencies you see in the dataset

a)

Data Type				The data types found in this dataset are Nominal, Ordinal, Interval and Ratio, as shown in the table to the left.
Not Numeric		Numeric		
Nominal	Ordinal	Interval	Ratio	
Make	Fuel	NormalizedLoss	RiskRating	
	Aspiration	Weelbase	EngSize	
	Doors	Length	Bore	
	Body	Width	Comp.ratio	
	Drive	Hight	HP	
	EngineLoc	Weight	PeakRPM	
	Engine Type	Stroke		
	Cylinder	CityMpg		
	FuelSys	HwyMpg		
		Price		

b)

Unique Value - Numeric						These are the unique values. Each value occurs only once in its attribute section, and this is what makes it unique.
NormalizedLoss	Weelbase	HP	Width	Stroke	CityMpg	
234.778	120.9	288	60.3	2.452	45	
256		264				
Unique Value – Not Numeric					These are the unique values (not numeric). Each value occurs only once in its attribute section, and this is what makes it unique.	
Make	Engine Type	Cylinder	FuelSys			
mercury	dohcv	three	mfi			
		twelve	spfi			

c)

Range - Numeric							
NormalizedLoss	Weelbase	Length	Width	Hight	Weight	Stroke	CityMpg
65 to 256	86.6 to 120.9	141.1 to 208.1	60.3 to 72.3	47.8 to 59.8	1488 to 4066	2.07 to 4.17	13 to 49
Range - Numeric							
RiskRating	EngSize	Bore	Comp.ratio	HP	PeakRPM	HwyMpg	Price
-2 to 3	61 to 326	2.54 to 3.94	7 to 23	48 to 288	4150 to 6600	16 to 54	5118 to 45400

d)

Categorical/Nominal – Enumerating all values for Make

MAKE === Confusion Matrix ===

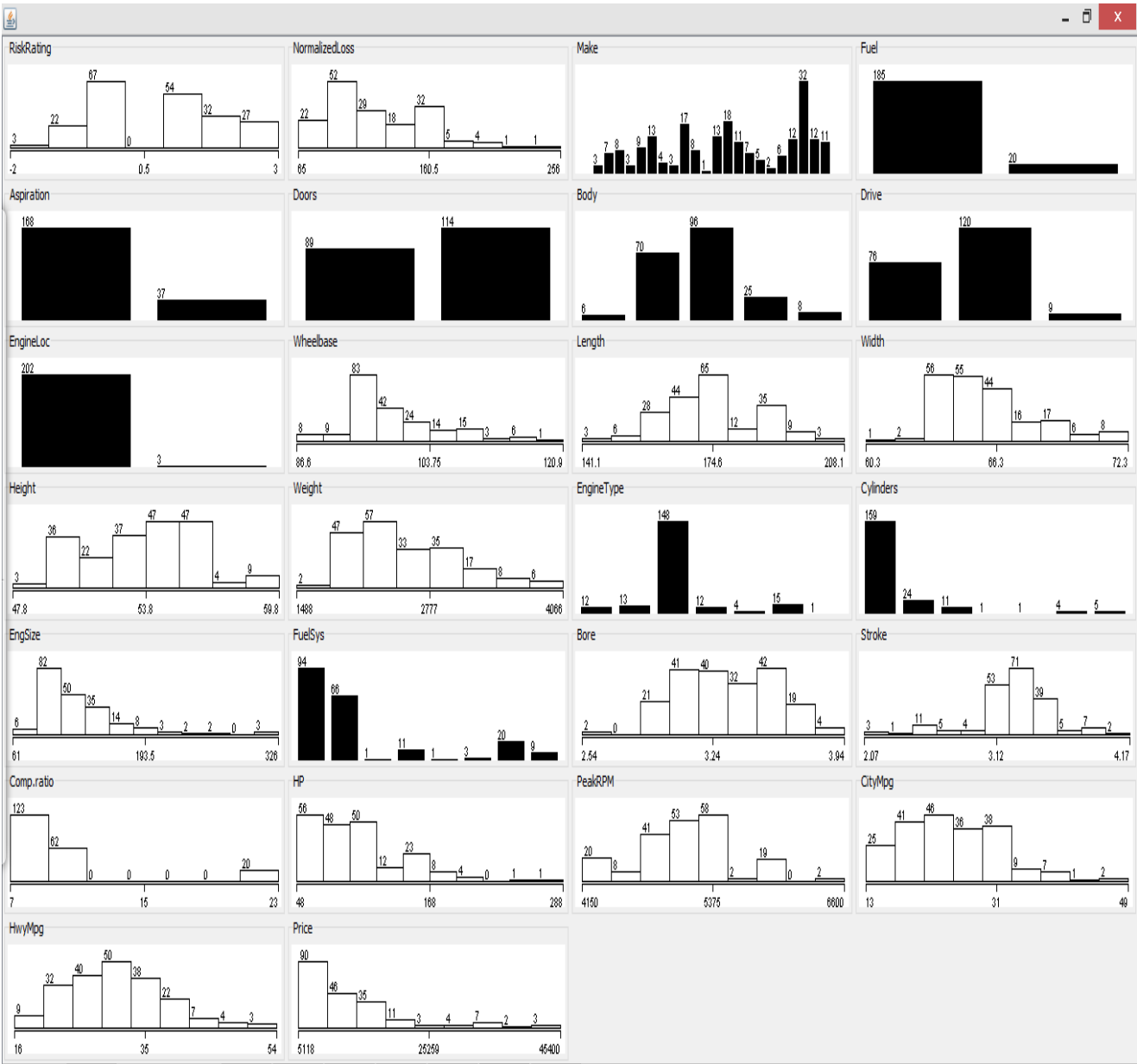
a b c d e f g h i j k l m n o p q r s t u v <-- classified as
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 a = alfa-romero
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 7 0 0 b = audi
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 8 0 0 c = bmw
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 d = chevrolet
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 9 0 0 e = dodge
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 13 0 0 f = honda
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 4 0 0 g = isuzu
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 h = jaguar
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 17 0 0 i = mazda
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 8 0 0 j = mercedes-benz
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 k = mercury
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 13 0 0 l = mitsubishi
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 18 0 0 m = nissan
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 11 0 0 n = peugot
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 7 0 0 o = plymouth
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 5 0 0 p = porsche
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 q = renault
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 6 0 0 r = saab
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 12 0 0 s = subaru
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 32 0 0 t = toyota
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 12 0 0 u = volkswagen
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 11 0 0 v = volvo

Ordinal – Enumerating all values for all cardinal attributes

FUEL === Confusion Matrix	ASPIRATIONS === Confusion Matrix	DOOR === Confusion Matrix	ENGINELOC === Confusion Matrix
a b <-- classified as	a b <-- classified as	a b <-- classified as	a b <-- classified as
185 0 a = gas	168 0 a = std	0 89 a = two	202 0 a = front
20 0 b = diesel	37 0 b = turbo	0 114 b = four	3 0 b = rear

DRIVE === Confusion Matrix =	BODY === Confusion Matrix ===	CYLINDER === Confusion Matrix ===
a b c <-- classified as	a b c d e <-- classified as	a b c d e f g <-- classified as
0 76 0 a = rwd	0 0 6 0 0 a = convertible	159 0 0 0 0 0 0 a = four
0 120 0 b = fwd	0 0 70 0 0 b = hatchback	24 0 0 0 0 0 0 b = six
0 9 0 c = 4wd	0 0 96 0 0 c = sedan	11 0 0 0 0 0 0 c = five
	0 0 25 0 0 d = wagon	1 0 0 0 0 0 0 d = three
	0 0 8 0 0 e = hardtop	1 0 0 0 0 0 0 e = twelve
		4 0 0 0 0 0 0 f = two
		5 0 0 0 0 0 0 g = eight

FUELSYS	ENGINE TYPE
=== Confusion Matrix ===	=== Confusion Matrix ===
a b c d e f g h <-- classified as	a b c d e f g <-- classified as
94 0 0 0 0 0 0 0 a = mpfi	0 0 12 0 0 0 0 a = dohc
66 0 0 0 0 0 0 0 b = 2bbl	0 0 13 0 0 0 0 b = ohcv
1 0 0 0 0 0 0 0 c = mfi	0 0 148 0 0 0 0 c = ohc
11 0 0 0 0 0 0 0 d = 1bbl	0 0 12 0 0 0 0 d = l
1 0 0 0 0 0 0 0 e = spfi	0 0 4 0 0 0 0 e = rotor
3 0 0 0 0 0 0 0 f = 4bbl	0 0 15 0 0 0 0 f = ohcf
20 0 0 0 0 0 0 0 g = idi	0 0 1 0 0 0 0 g = dohcv
9 0 0 0 0 0 0 0 h = spdi	



e)

Inconsistencies						
	RiskRating	Bore	Stroke	HP	PeakRPM	Price
Unknown Value (?)	41	4	4	2	2	4
Ignored Class	41	4	4	2	2	4

The inconsistencies that I found in the dataset was the fact that some attributes, such as those shown above, have values that are unknown, such as “?”. This symbol cannot be identified as any of the data type found in the dataset, and therefore it causes issues when processing the data. The table above shows the attributes that contain this symbol, and the number of time that is being repeated in that column. If you observe the output generated below, this values were ignored because they do not add any information to the dataset.

RISK RATING

=== Evaluation on training set ===

=== Summary ===

Correlation coefficient 0
Mean absolute error 29.122
Root mean squared error 35.3339
Relative absolute error 100 %
Root relative squared error 100 %
Total Number of Instances 64
Ignored Class Unknown Instances 41

BORE

Correlation coefficient 0
Mean absolute error 0.2358
Root mean squared error 0.2729
Relative absolute error 100%
Root relative squared error 100 %
Total Number of Instances 201
Ignored Class Unknown Instances 4

STROKE

Correlation coefficient 0
Mean absolute error 0.2318
Root mean squared error 0.3159
Relative absolute error 100 %
Root relative squared error 100 %
Total Number of Instances 201
Ignored Class Unknown Instances 4

HP

=== Evaluation on training set ===

=== Summary ===

Correlation coefficient 0
Mean absolute error 30.5049
Root mean squared error 39.6164
Relative absolute error 100 %
Root relative squared error 100 %
Total Number of Instances 203
Ignored Class Unknown Instances 2

PEAK RPM

Correlation coefficient 0
Mean absolute error 385.8283
Root mean squared error 478.1525
Relative absolute error 100 %
Root relative squared error 100 %
Total Number of Instances 203
Ignored Class Unknown Instances 2

PRICE

Correlation coefficient 0
Mean absolute error 5841.1883
Root mean squared error 7927.2729
Relative absolute error 100 %
Root relative squared error 100 %
Total Number of Instances 201
Ignored Class Unknown Instances 4